Protecting Salem's Ecosystem
Predictive Modeling of Stream Temperature for the City of Salem

Serenna Walter · Siera Edwards · Brooke Proctor
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Abstract

Rising stream temperatures are threatening Salem's ecosystem. The loss in tree canopy coverage and increasing temperatures risk pushing native species to cooler bodies of water and away from Salem's waterways. This project models the change in Salem's Mill Creek temperature from 2015 to 2025 using historical stream temperature, weather, and canopy coverage data using linear regression models. One model stood out among the rest, performing exceptionally well. This model combines interactions within temperature and time to predict stream temperatures. Within the model, with the available data, tree canopy does not have a significant impact on stream temperature. The main driver of the model is time. This means that, as it stands, more work should be done in protecting stream temperature against future rising temperatures, regardless of tree canopy coverage. A defining challenge in this project included a lack of reliable data; thus, it is also recommended that the City of Salem allocate more resources to collecting data that would shed more light on factors impacting stream temperature.

Introduction & Motivation

Problem Statement:
The City of Salem is concerned about increasing stream temperatures and the impact that this will have on the environment [4]. Stream temperature rise can impact Indigenous communities who rely on Salmon populations, citizens who utilize streams for cooling or drinking, and anyone else who uses water in Salem [4]. Additionally, rising stream temperatures threaten local species and spur harmful algal blooms, posing serious risks to the health of Salem's ecosystem.

Research Questions:
1. Can historic stream temperature data be used to predict stream temperature with an R-squared value of at least 90%?
   a. Will adding historic weather data for Salem, Oregon improve the R-squared value from the initial model?
   b. Will adding historic tree canopy data for Salem, Oregon improve the R-squared value from the initial model?
2. Can a model be used to predict Salem's stream temperature 3 years into the future based on different levels of anticipated climate change impact on the weather data?

Background & Related Work

Bodies of water around the world are beginning to heat up at faster rates than air heat waves [1]. These water heat waves–defined by at least five consecutive days above the seasonal average–result in lower dissolved oxygen content, nutrient cycling, and pH buffering [2]. All of this means that ecosystems and native species adapted to live in colder waters face serious threats, while algae has room to become prolific.

One way to combat high stream temperatures is to support riparian zones–the dense plant areas along rivers [3]. These areas often sport trees and other foliage that shade and cool rivers and streams, effectively lowering the water temperature. However, these areas are at risk for several reasons. Some of these reasons are forest clearing, runoff from urban and agricultural lands, and human development [3].

This recent expansion of threats to river temperatures has caused Oregon to start thinking critically about readiness and response plans, including Salem's own plan [4]. Limited research currently exists on stream temperatures, weather, and tree canopies in Salem. This research will support the City of Salem in addressing critical preparation gaps for warmer temperatures and inform stream cooling efforts. This research could also extend to support private organizations in the prevention of the loss or disruption of riparian zones in Salem.

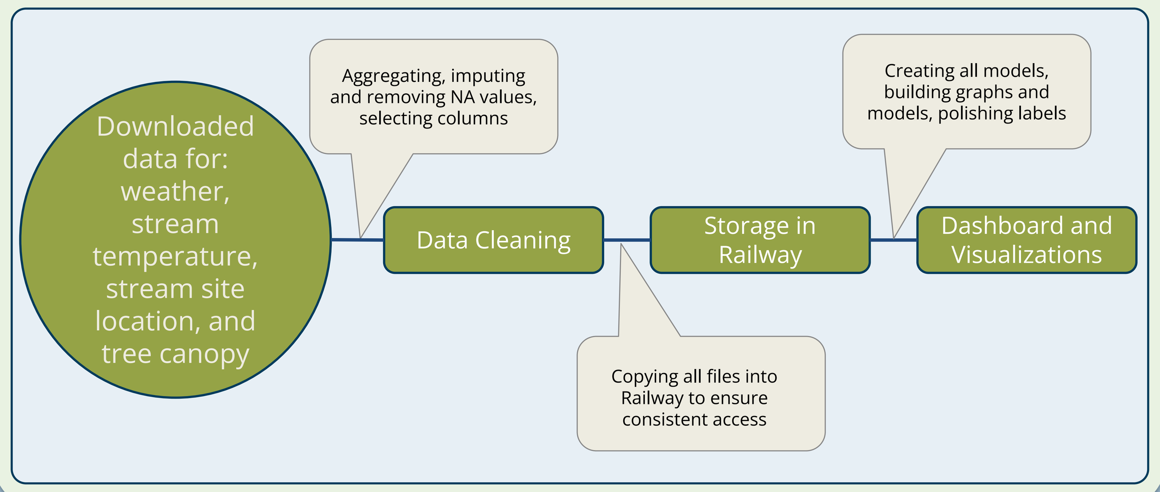
Data, Engineering & Ethics

Sources & Scale:
The data is sourced from NOAA, The City of Salem, Tree Plotter Canopy, and CastawayWU. These were all available with permission from the data publishers, and could be cleanly downloaded as a flat CSV file.

Ethics & Responsible Use:
The data contains no PII, so no personal identifiable information is at risk. A main ethical concern with this project is creating predictions with the limited years of data that we have access to. This risks missing overall trends that are seen across years. This also limits the ability to make predictions into the future, instead we must focus on emphasising the scope of stream temperature change in Salem between years.

Stream Sites:
This used two site locations in Salem: MIC3 and MIC12. MIC3 is further north and has more tree canopy, whereas MIC12 is further south and gets more sun. Both sites are part of Mill Creek, flowing west into the Willamette River.

Pipeline:

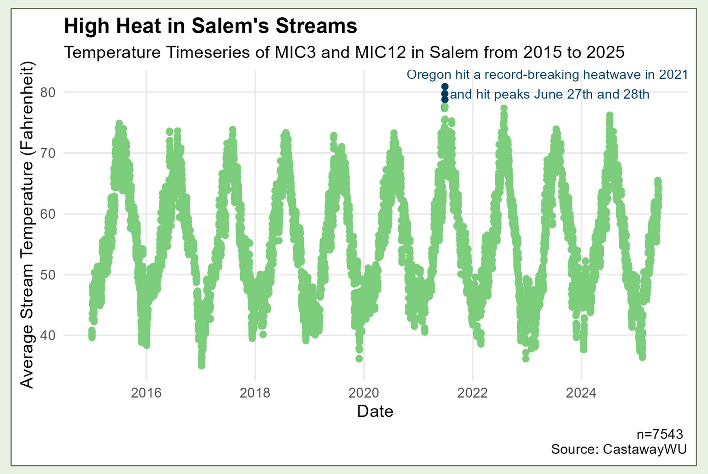


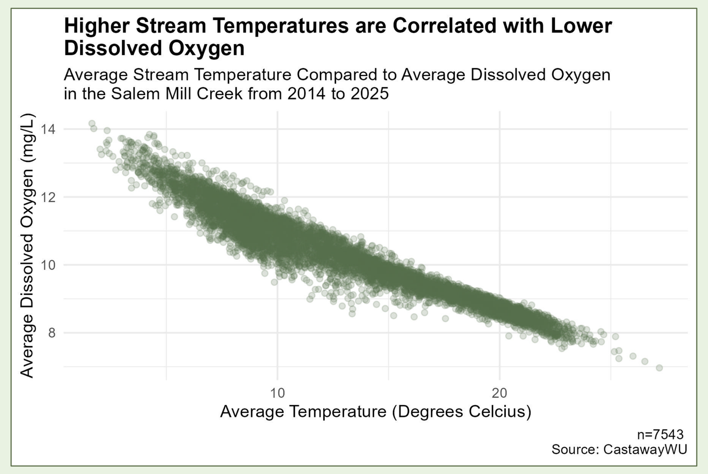
Methods & Evaluation

Approach:
To answer the research questions, linear regression models with interactions between features were created in R iteratively, with different data included for each model. The linear regression models predicting stream temperature were done in the following order:
1. Using 2015-2025 stream temperature data only
2. Using 2015-2025 stream temperature and weather data
3. Using 2020, 2022, and 2024 stream temperature, weather, and tree canopy data

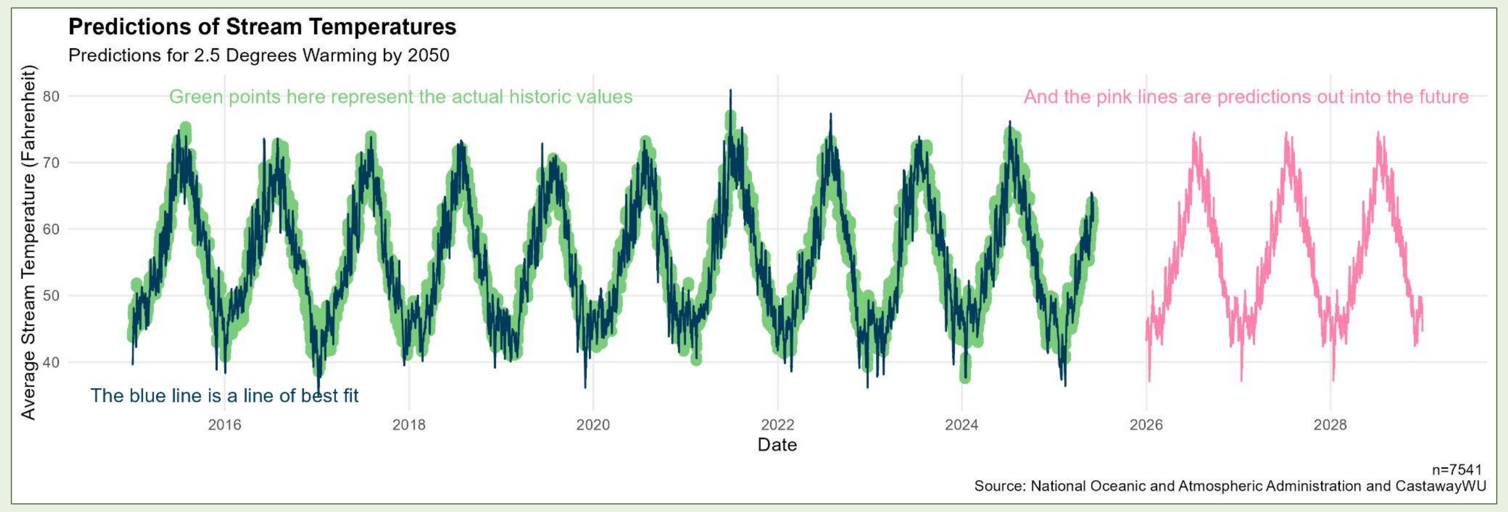
Validation:
The function plot() was utilized to check model assumptions. R-Squared, P-values, Root Mean Squared Error (RMSE), and Residual Standard Error to evaluate model performance.

Results & Visualizations





The highest performing model was a linear regression model that used stream temperature data and weather data, but did not include tree canopy data. When fitting this model, each year was used as a different factor level and there were interactions between correlated independent variables. This model produces an adjusted R-squared value of 0.9442. The model was adapted slightly, then used to predict stream temperature values for three years in the future (2026-2028).



Discussion, Limitations & Conclusions

Discussion:
This project shows that historic stream temperature data can be used to predict stream temperature with a high R-squared value of 90.0%, according to the best model in this project. The inclusion of weather data increases this model's performance with an R-squared value of 94.5%. The addition of tree canopy data increases the model's performance with an R-squared value of 94.7%, but since this data has an extremely limited range for year, it is difficult to tell if the addition of the tree canopy data truly would improve the model over the same period of time as the other models, or if its increased R-squared value is simply due to less observations.

Limitations & Future Work:
Linear regression models assume that there is no multicollinearity between independent variables. Unfortunately, there are multiple variables in the weather dataset that are known to be highly correlated. The chosen model accounts for the correlation between variables, but there are still risks that the model is underrepresenting the significance of any highly correlated variables. Future work could extend this project by the inclusion of data on tree canopy yearly, stream temperature for all streams across Salem outside of Mill Creek, the impact of invasive species along streams, and snowfall/snowmelt. Future work could also include a seasonal ARIMA model for more granularity and accuracy in future predictions.

Conclusions:
This study shows that historic stream temperature data and weather data can predict stream temperature in Mill Creek, with the best model explaining 94.5% of the variance and an RMSE of 2.11°F. This study also shows that tree canopy has the possibility of positively impacting the model, but the current ability is extremely limited due to the number of years present in the data. Further research is needed to determine the impact of tree canopy on predicting stream temperature.

Reproducibility & References

Repository: https://github.com/Serenna4/Emerald-Ash-Borer-Capstone-Project/tree/main
Key dependencies: R 4.5.2, Beekeeper SQL 5.9, Railway | Libraries: tidyverse, gridExtra, car, forecast

References:
[1] Bush, E., U.S. rivers are experiencing unprecedented and unexpectedly intense warming, NBC News, 2025.
[2] Penn State Extension, We are in Hot Water–Streams and Rivers are Heating Up, Pennsylvania State University, 2026
[3] University of Georgia, Stream Warming and Brook Trout in the Southeast, University of Georgia, 2026.
[4] City of Salem, City of Salem Emerald Ash Borer Readiness and Response Plan, City of Salem, 2026.

Acknowledgements

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